RESEARCH REPORT

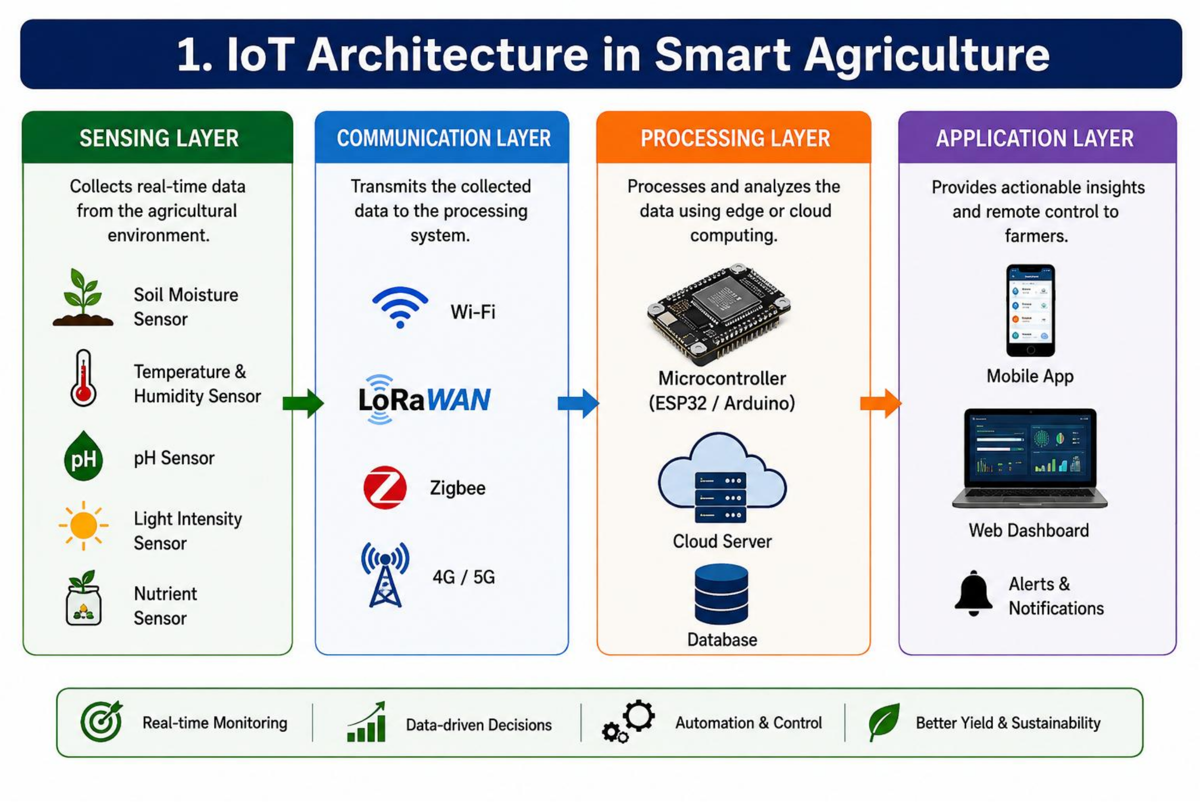
IoT in Real Life: Applications in Smart Agriculture

Introduction

The Internet of Things (IoT) is a technology in which physical devices such as sensors, machines, controllers, and appliances are connected to the Internet so that they can collect, exchange, and process data. IoT has become increasingly important in agriculture because traditional farming depends heavily on manual observation and experience. Farmers often need to make decisions about irrigation, fertilizers, crop health, and environmental conditions without having continuous information about the field.

Smart Agriculture, also known as precision agriculture, uses IoT devices to collect real-time information about soil, crops, weather, water, and other agricultural conditions. This information can then be analyzed and used to automate farming operations. According to the Food and Agriculture Organization (FAO), digital technologies can help farmers improve productivity, resource efficiency, and environmental sustainability. [1], [2]

1. IoT Architecture in Smart Agriculture

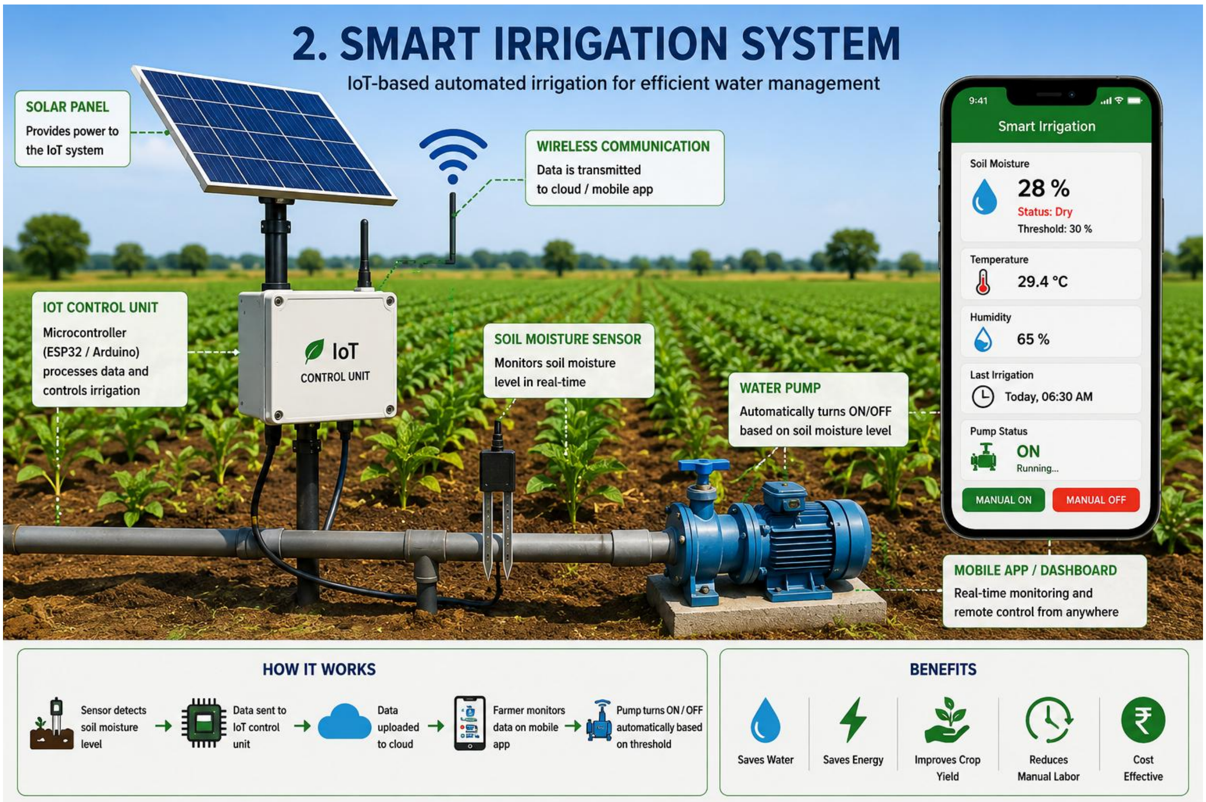


A typical IoT-based smart agriculture system consists of four major layers:

1. **Sensing Layer:** Sensors collect information from the agricultural environment. Common sensors measure soil moisture, temperature, humidity, pH, light intensity, water level, and nutrient conditions.
2. **Communication Layer:** The collected data is transmitted using technologies such as Wi-Fi, Bluetooth, Zigbee, cellular networks, or LoRa/LoRaWAN.
3. **Processing Layer:** A microcontroller such as Arduino or ESP32, together with cloud or edge computing platforms, processes the sensor information.
4. **Application Layer:** Farmers can view information through a mobile application or web dashboard and control agricultural equipment remotely.

This architecture allows farming systems to move from manual decision-making toward real-time, data-driven management.

2. Smart Irrigation



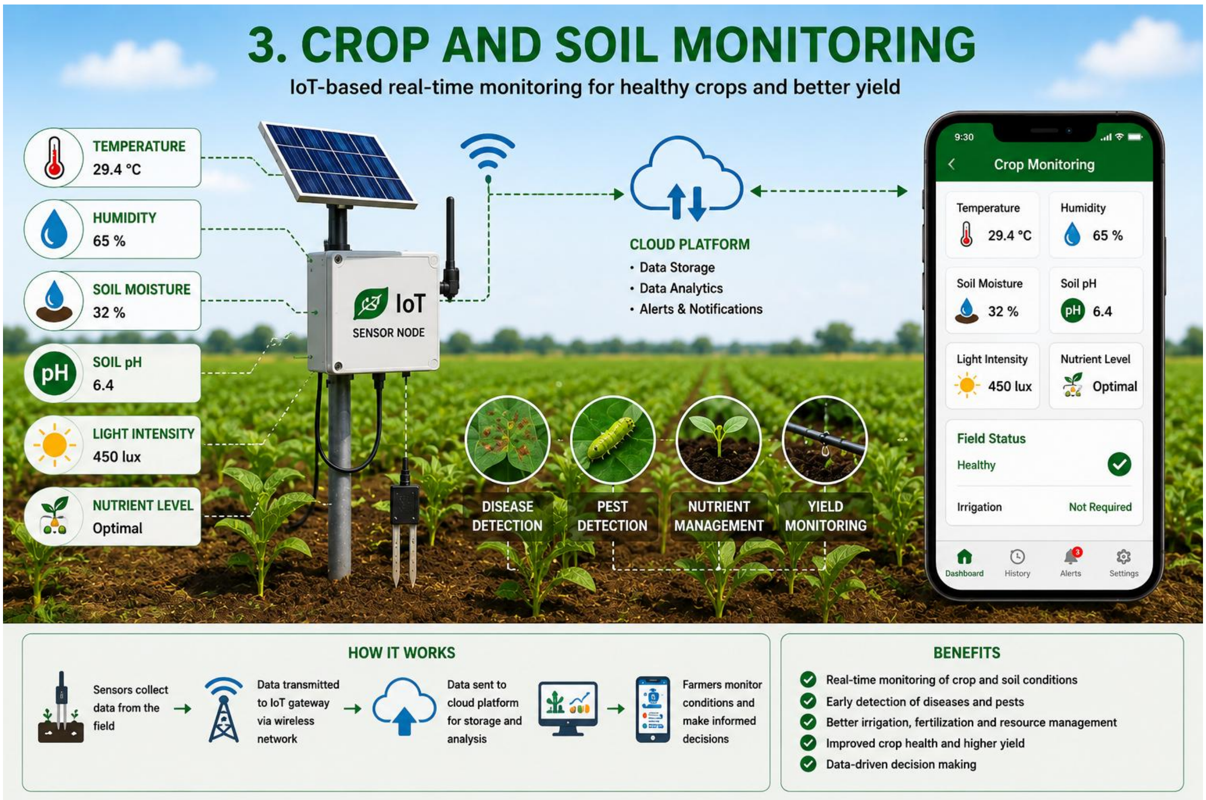
One of the most important real-life applications of IoT in agriculture is **smart irrigation**. Conventional irrigation can waste water because crops may be irrigated according to a fixed schedule rather than their actual water requirements.

In an IoT-based irrigation system, soil-moisture sensors continuously measure the amount of water present in the soil. When the moisture level falls below a predefined threshold, the controller can automatically activate a water pump. When sufficient moisture is detected, the pump is switched off.

More advanced systems combine soil moisture data with temperature, humidity, rainfall, and weather forecasts. IoT-based irrigation research has shown significant potential for reducing unnecessary water consumption while maintaining suitable soil moisture for crops. [3], [4]

For example, a farmer can monitor soil moisture through a smartphone and switch irrigation equipment on or off remotely. This reduces water wastage, electricity consumption, and manual labour.

3. Crop and Soil Monitoring



IoT sensors can continuously monitor the condition of crops and soil. Sensors can measure parameters such as soil moisture, temperature, humidity, pH, electrical conductivity, and nutrient levels.

This information helps farmers determine whether a crop is receiving suitable growing conditions. If abnormal temperature, moisture, or other environmental conditions are detected, the system can generate an alert.

IoT can also be combined with cameras, drones, and artificial intelligence to identify crop diseases, pest attacks, weeds, and plant stress. Research has identified crop monitoring, disease detection, fertilizer management, climate monitoring, and yield monitoring as important agricultural IoT applications. [5]

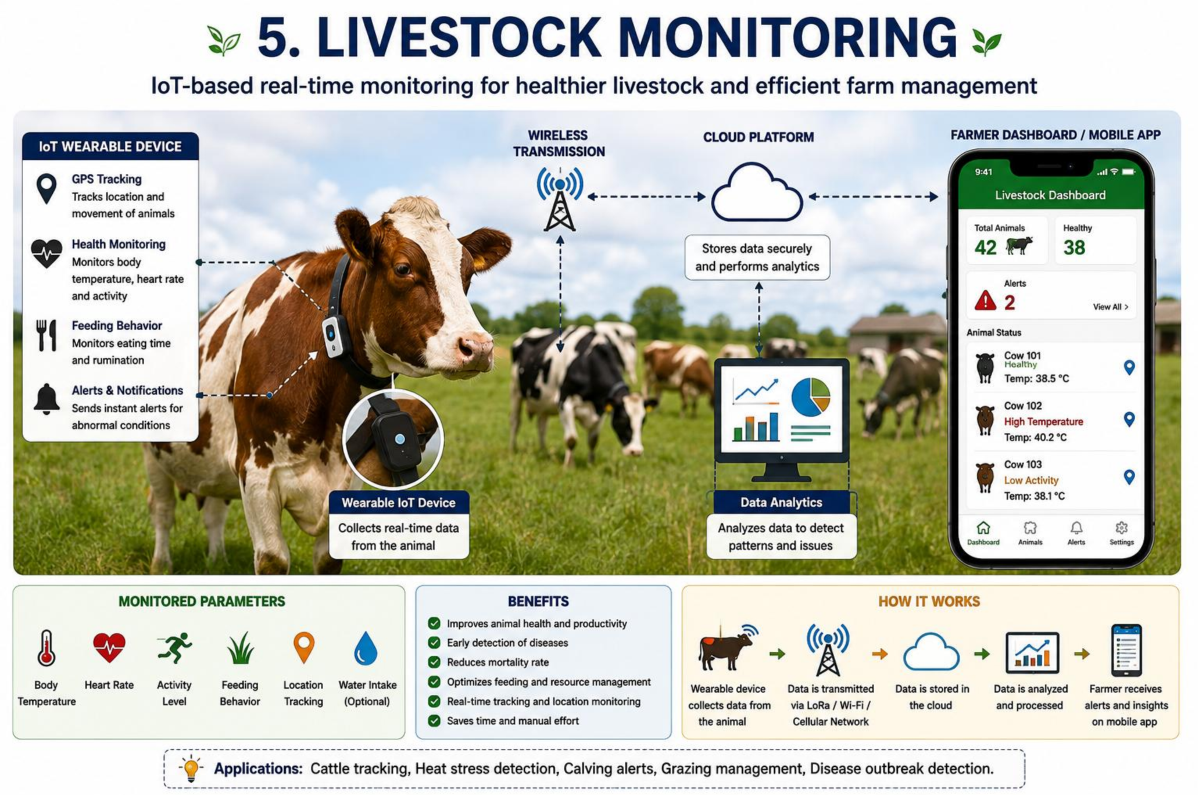
4. Smart Greenhouses

IoT is also widely applicable to greenhouse farming. A greenhouse requires controlled environmental conditions for healthy plant growth. Sensors can monitor temperature, humidity, light intensity, soil moisture, and carbon dioxide levels.

Based on sensor readings, actuators can automatically control fans, irrigation pumps, lights, ventilation systems, and other equipment. For example, if the temperature becomes too high, the controller can activate ventilation fans. Similarly, if soil moisture decreases, an automated irrigation system can supply water.

This approach reduces continuous human supervision and provides plants with more consistent growing conditions.

5. Livestock Monitoring



IoT is not limited to crops. It can also be used for **livestock management**. Wearable sensors and connected devices can monitor animal location, movement, body temperature, feeding behaviour, and other indicators.

If an animal shows unusual movement or abnormal temperature, the farmer can receive an alert and investigate the animal. GPS-enabled devices can also help farmers track livestock and reduce the risk of animals becoming lost.

6. Benefits of IoT in Agriculture

The major benefits of IoT-based agriculture include:

- **Efficient water usage:** Irrigation can be based on actual soil conditions.
- **Reduced labour:** Automatic monitoring and control reduce manual work.
- **Improved crop productivity:** Farmers can respond quickly to unsuitable conditions.
- **Better resource management:** Fertilizers, water, energy, and pesticides can be applied more precisely.
- **Real-time monitoring:** Farmers can access field information remotely.
- **Early detection:** Diseases, pests, and environmental problems can be identified earlier.
- **Data-driven decision-making:** Historical and real-time data can help farmers make better decisions.

FAO notes that IoT can help farmers optimize agricultural inputs, reduce production costs, improve crop monitoring, and optimize water use. [2]

7. Challenges

Despite its advantages, IoT adoption in agriculture faces several challenges. The initial cost of sensors, controllers, communication equipment, and software can be difficult for small-scale farmers. Rural areas may also have limited Internet or cellular connectivity.

Other challenges include sensor maintenance, battery life, data security, interoperability between different IoT devices, and the need for technical knowledge. FAO research has identified high investment costs, limited digital skills, and inadequate supporting infrastructure as important barriers to agricultural digitalization. [6]

Therefore, affordable IoT devices, low-power communication technologies, farmer training, and reliable rural connectivity are essential for widespread adoption.

8. Future Scope

The future of smart agriculture will involve greater integration of **IoT, Artificial Intelligence (AI), Machine Learning (ML), drones, satellite imagery, robotics, and cloud computing**. AI can analyze large amounts of sensor data and predict irrigation requirements, crop diseases, weather-related risks, and crop yields.

Low-power technologies such as LoRaWAN can enable sensors to communicate over long distances while consuming relatively little energy. Combining IoT with AI can therefore create autonomous agricultural systems capable of monitoring fields and making recommendations with minimal human intervention. [3], [7]

9. Conclusion

IoT is transforming agriculture from a traditional, experience-based activity into a more precise and data-driven process. Smart irrigation, soil and crop monitoring, greenhouse automation, livestock tracking, and disease detection are practical examples of IoT being used in real life.

Although challenges such as cost, connectivity, maintenance, and technical skills remain, IoT has significant potential to improve agricultural productivity while reducing the consumption of water, energy, fertilizers, and other resources. With the integration of AI, drones, cloud computing, and advanced sensors, IoT-based smart agriculture can play an important role in building a more efficient and sustainable agricultural future.

References

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[3] B. Nsoh *et al.*, “Internet of Things-Based Automated Solutions Utilizing Machine Learning for Smart and Real-Time Irrigation Management: A Review,” *Sensors*, vol. 24, no. 23, 2024, Art. no. 7480.

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[7] “A Comprehensive Review on Smart and Sustainable Agriculture Using IoT Technologies,” *Smart Agricultural Technology*, 2024.